

# Stage 1 — The problem, the pilot and the protocol

Where the Shortcut Sits: supervisor meeting 1 of 6

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**In one sentence.** Region-of-interest (ROI) masking is the standard defence against acquisition artifacts that act as shortcuts; this project asks whether it still works when the artifact lies *inside* the ROI. This first meeting presents the clinical problem, the pilot that raised the question, its independent replication, and the protocol that every later stage uses — including a correction to the bootstrap that widened every interval.

## Notation

|                  |                                                                                                                                                                                                          |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $Y \in \{0, 1\}$ | diagnosis (1 = melanoma / malignant / solid-component tumour / erosion).                                                                                                                                 |
| $A \in \{0, 1\}$ | artifact present (hair, sonographer caliper, capsule debris, chest tube).                                                                                                                                |
| ROI              | region of interest (lesion, nodule, tumour, lesion box, lungs); drawn in <b>green</b> ; artifacts in <b>red</b> .                                                                                        |
| $r$              | overlap $ \text{artifact} \cap \text{ROI} / \text{artifact} $ . Trap A: $r \geq 0.5$ (in-ROI); Trap B: $r < 0.1$ (out-of-ROI).                                                                           |
| Environments     | training and correlated test $P(A=1   Y=1) = 0.9$ , $P(A=1   Y=0) = 0.1$ ; reversed test 0.1/0.9; clean test: artifact independent of $Y$ (real traps: half of each class; sweeps: absent).              |
| ERM, mask        | linear head on the frozen features of the original image / of the ROI-masked image.                                                                                                                      |
| Crossover        | $[\text{mask} - \text{ERM}]_{\text{out}} - [\text{mask} - \text{ERM}]_{\text{in}}$ of reversed-test AUROC (real traps: Trap B – Trap A; sweeps and transplant: location interaction on the same images). |
| CI               | 95% crossed seed×image bootstrap interval (Stage 1, Sec. 3.1); every interval in these reports uses it.                                                                                                  |
| $ \Delta p $     | same-head counterfactual sensitivity: change of the predicted probability when the artifact is inserted.                                                                                                 |
| SMD              | standardised mean difference between two groups.                                                                                                                                                         |

## 1 Where we are

This is the first of six meetings. It sets up everything the later stages rely on: the question, the evidence from the pilot study (experiments E1–E15), an independent re-implementation of that pilot from its written specification, and the experimental protocol — environments, frozen backbones, linear heads, grouped splits, pre-registration and the statistical estimator. What is new relative to the pilot is (i) that every pilot claim was re-derived from scratch and compared number by number, and (ii) that the uncertainty estimator was corrected after a flaw was found late in the project; all intervals in these six reports use the corrected estimator.

## 2 Question

### 2.1 Why shortcut artifacts matter clinically

Deep classifiers exploit features that are correlated with the diagnosis in the training data but are not evidence of disease (Geirhos et al., 2020; DeGrave et al., 2021). In dermoscopy, surgical skin markings (Winkler et al., 2019) and ruler scale bars (Winkler et al., 2021) changed the melanoma predictions of a market-approved system; in chest radiography, pneumothorax classifiers detect the chest drain that treats the condition (Oakden-Rayner et al., 2020; Jiménez-Sánchez et al., 2023). Such correlations arise from clinical workflow — suspicious lesions are measured and marked, sonographers place calipers on nodules they consider suspicious — so they are present in every retrospectively collected dataset, and a model that relies on them fails exactly on the patients whose workflow differs (for example, a malignant nodule that was not measured).

### 2.2 The research question and why we asked it this way

The most widely recommended defence is to restrict the classifier to the ROI: segment the lesion and discard the background. Its rationale assumes that the shortcut lives in the background. Rulers, hair and calipers do not respect that assumption; when an artifact overlaps the ROI, masking cannot remove it and removes everything *else*. The question is therefore:

*Does the benefit of ROI masking depend on where the artifact sits relative to the ROI — and if so, which quantity governs that dependence, and does it matter on unaltered clinical data?*

### Design decisions and rejected alternatives

- *Control the overlap first, then use real artifacts.* The pilot pasted a synthetic ruler at controlled overlap with the lesion (0–100%). This isolates location from everything else; real artifacts (Stage 3) come later because their location is confounded with the images that carry them (Stage 4). Starting with real artifacts only was rejected because a location effect could not be separated from population differences.
- *Frozen foundation-model features and linear heads.* This is how foundation models are typically deployed on small medical datasets, it makes every arm comparable (same features, only the head or the input changes), and it is cheap enough to repeat over seeds, overlaps and cohorts. End-to-end fine-tuning was kept as a replication check (Stage 5), not the main protocol, because it multiplies compute by two orders of magnitude.
- *A trap design with a reversed test.* Training with a strong artifact–label association and testing on the reversed association measures shortcut reliance directly; clean and correlated tests measure the cost of a mitigation. A single natural test set was rejected as the main design because its shortcut is too weak to measure location effects precisely; natural test sets are used in Stage 5.

## 3 Methods

**Pilot cohort** ISIC 2018 Task 1/2 images with manual lesion masks and melanoma labels from the union of the ISIC 2016 and 2017 tables (three images with conflicting labels dropped); perceptual-hash (Hamming  $\leq 8$ ) and lesion grouping; 2,437 images for which a  $65 \times 19$  px ruler can be placed at 0/25/50/75/100% overlap within 10 percentage points (Codella et al., 2019). Placements are frozen and reused by every run.

**Backbones and heads** DINOv2 ViT-B/14 at 518px (Oquab et al., 2024) (CLS token), DINOv2 at 224px and the dermatology foundation model DermLIP at 224px (Yan et al., 2025); logistic-regression heads; regularisation and decision threshold (maximal balanced accuracy) chosen on clean validation data only.

**Environments** Training and correlated test use  $P(A=1 | Y=1) = 0.9$  and  $P(A=1 | Y=0) = 0.1$ ; the reversed test swaps them; the clean test has no artifact (sweeps) or an artifact independent of the label (real traps). Figure 1 shows the four environments. Seeds: 42, 123, 456 for the sweeps.

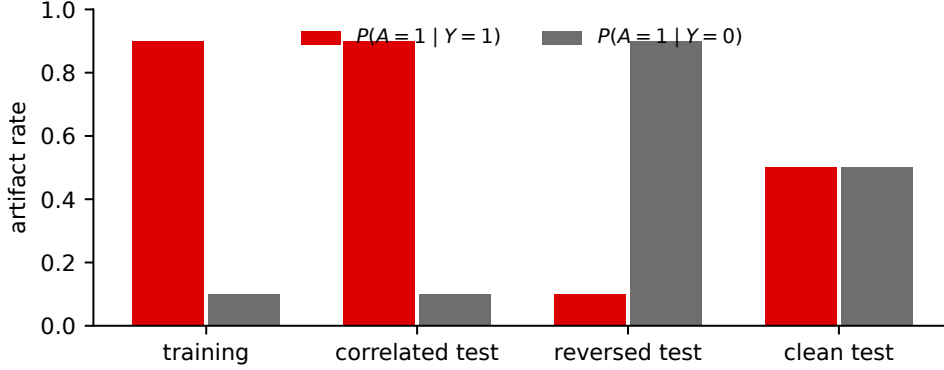


Figure 1: The trap environments. A model that uses the artifact gains on the correlated test and collapses on the reversed one; the reversed-test AUROC is the primary metric.

**Splits and leakage** Splits are grouped by lesion identifier and perceptual hash; a leakage experiment (Sec. 4.3) compares grouped with random splits. Table 1 summarises the protocol.

Table 1: The protocol used unchanged in every later stage.

| Component      | Choice                                                                                                                                    | Why                                                                      |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
| Environments   | training and correlated test $P(A=1   Y=1) = 0.9$ , $P(A=1   Y=0) = 0.1$ ; reversed test swapped; clean test artifact-free or independent | a strong, known shortcut whose use is measurable by its reversal         |
| Primary metric | reversed-test AUROC; clean and correlated AUROC as secondary                                                                              | threshold-free; reversal isolates shortcut use                           |
| Encoders       | frozen DINOv2 ViT-B/14 at 518 px (all cohorts); DermLIP, MedSigLIP, RAD-DINO, ConvNeXt, DINOv2 S/L as replications                        | typical deployment on small medical data; identical features across arms |
| Heads          | logistic regression; C and threshold chosen on clean validation data only                                                                 | no test information in model selection                                   |
| Arms           | ERM; ROI masking; oracle inpainting; group balancing; DFR; LEACE; GroupDRO; later remedies                                                | masking is compared with remedies that target the artifact               |
| Splits         | grouped by lesion identifier, perceptual hash, patient where available                                                                    | no near-duplicate leakage (Sec. 4.3)                                     |
| Seeds          | 3 seeds (sweeps); 5 seeds $\times$ 5 grouped folds (real traps)                                                                           | variation of training and split                                          |
| Uncertainty    | crossed seed $\times$ image bootstrap, 10,000 replicates                                                                                  | seeds share test images (Sec. 3.1)                                       |
| Registration   | design, claims and support criteria committed before each experiment                                                                      | separates confirmation from exploration                                  |

### 3.1 The corrected bootstrap

Every comparison is a paired difference of AUROCs between two arms on the same test images, averaged over training seeds. The original estimator (inherited from the pilot) resampled seeds and then, *independently within each sampled seed*, the test images. In most analyses, however, every seed scores the *same* test images (a fixed test split, or cross-validation folds that together cover the same pool). Resampling images independently per seed then treats one test set as several and divides the test-sampling variance by the number of seeds: intervals were too narrow. The

corrected (*crossed*) estimator resamples seeds with replacement and draws one Poisson(1) weight per test image, shared by every seed in which the image appears, then recomputes the per-seed statistic and averages (`wtss.stats_crossed`; 10,000 replicates). Every interval in these reports comes from regenerated runs with this estimator (Sec. 5.1 gives its consequences).

### 3.2 Pre-registration

Before every experiment a document stating the design, the claims and what would count as support was committed to the repository; results were committed afterwards. The registry lists 23 registrations; every one precedes its first outcome result except that the SLAS registration and its localisation result share a commit, and the 16-test primary family (Stage 3) was grouped after the individual results. **Caveat:** commit times come from the machine that made the commits and are not independent proof of order; only the registrations of the last three rounds were additionally pushed to GitHub before running, which provides an external time stamp.

## 4 Experiments

### 4.1 The pilot (E1–E15)

post hoc (the pilot preceded this project’s pre-registration; its specification was written down afterwards and then re-implemented). Table 2 lists what each pilot experiment tested.

Table 2: The pilot experiments (specification: `docs/REPLICATION_SPEC.md`).

| Exp. | What it tests                                                                               |
|------|---------------------------------------------------------------------------------------------|
| E1   | controlled ruler sweep, DINOv2@518: mask – ERM at 0–100% overlap; location interaction      |
| E2   | the same at 224 px (DINOv2) and with the dermatology model DermLIP                          |
| E3   | appearance randomisation (colour, opacity, ticks, scale of the ruler)                       |
| E4   | occlusion control: ruler present but uncorrelated with the label (50/50)                    |
| E5   | mask dilation: restoring ruler pixels around the lesion (retention)                         |
| E6   | same-head counterfactual sensitivity $ \Delta p $ to inserting the ruler                    |
| E7   | lesion-size strata (artifact-to-lesion area ratio)                                          |
| E8   | oracle inpainting and a prediction-consistency repair                                       |
| E9   | bridge: group-balanced training, DFR, GroupDRO, paired LEACE at 100% overlap                |
| E10  | hair decodability from the embedding; paired LEACE on real hair                             |
| E11  | atlas of real artifacts (hair, ink, vignetting) in ISIC 2019 / HAM10000; vignetting trap    |
| E12  | overlap-contrast trap: in-lesion versus out-of-lesion hair, no artifact-free group          |
| E13  | real hair traps (Trap A in-lesion, Trap B out-of-lesion, shared hair-free group), ISIC 2019 |
| E14  | area-ratio strata of the real hair trap                                                     |
| E15  | E13 with DermLIP                                                                            |

### 4.2 Independent replication of the pilot

registered before results (specification `docs/REPLICATION_SPEC.md`, first commit 2699d51, 2026-09-26 15:47). The pilot was re-implemented from its written specification and every reported number was compared with the re-derived one. *Support* means: same sign, same verdict on whether the interval excludes zero, and  $|\Delta| \leq 0.02$  (MATCH); SIGN+CI if the sign and verdict agree but  $|\Delta| > 0.02$ ; MISMATCH otherwise.

### 4.3 Leakage: grouped versus random splits

registered before results (part of the replication specification). The controlled dermoscopy sweep was repeated with five random image-level splits instead of the grouped split, holding everything else fixed.

## 5 Results

**Replication** The original ledger (before the bootstrap correction) had 73 MATCH, 12 SIGN+CI and 3 MISMATCH among 88 numbers. Re-checked with the regenerated runs and the corrected estimator, the counts are 76 MATCH, 6 SIGN+CI and 5 MISMATCH among 87 numbers (Table 3; the pilot’s values are shown as point estimates only). The central pilot result replicates: in the controlled sweep, masking changes reversed-test AUROC by +0.184 [+0.117, +0.250] at 0% overlap and by −0.129 [−0.195, −0.064] at 100% overlap (location interaction +0.313 [+0.258, +0.370]).

Table 3: Pilot replication ledger, re-checked with the regenerated runs and the corrected estimator. The pilot’s own intervals (old estimator) are not reproduced.

| Exp. | Quantity                                 | Pilot (point) | Replication (crossed CI) | Verdict  |
|------|------------------------------------------|---------------|--------------------------|----------|
| E1   | mask – ERM rev @0%                       | +0.184        | +0.184 [+0.117, +0.250]  | MATCH    |
| E1   | mask – ERM rev @25%                      | -0.058        | -0.059 [-0.127, +0.008]  | MISMATCH |
| E1   | mask – ERM rev @50%                      | -0.097        | -0.097 [-0.164, -0.030]  | MATCH    |
| E1   | mask – ERM rev @75%                      | -0.111        | -0.111 [-0.181, -0.043]  | MATCH    |
| E1   | mask – ERM rev @100%                     | -0.128        | -0.129 [-0.195, -0.064]  | MATCH    |
| E1   | interaction 0% – 100% (mask)             | +0.312        | +0.313 [+0.258, +0.370]  | MATCH    |
| E2   | dino224 mask – ERM @0%                   | +0.364        | +0.361 [+0.275, +0.444]  | MATCH    |
| E2   | dino224 mask – ERM @25%                  | -0.040        | -0.049 [-0.117, +0.018]  | MATCH    |
| E2   | dino224 mask – ERM @50%                  | -0.076        | -0.081 [-0.143, -0.019]  | MATCH    |
| E2   | dino224 mask – ERM @75%                  | -0.102        | -0.106 [-0.166, -0.044]  | MATCH    |
| E2   | dino224 mask – ERM @100%                 | -0.092        | -0.095 [-0.156, -0.035]  | MATCH    |
| E2   | dermlip224 mask – ERM @0%                | +0.160        | +0.161 [+0.102, +0.222]  | MATCH    |
| E2   | dermlip224 mask – ERM @25%               | -0.291        | -0.289 [-0.366, -0.215]  | MATCH    |
| E2   | dermlip224 mask – ERM @50%               | -0.300        | -0.298 [-0.371, -0.225]  | MATCH    |
| E2   | dermlip224 mask – ERM @75%               | -0.313        | -0.309 [-0.378, -0.239]  | MATCH    |
| E2   | dermlip224 mask – ERM @100%              | -0.268        | -0.263 [-0.348, -0.180]  | MATCH    |
| E3   | variable ruler mask – ERM @0%            | +0.147        | +0.146 [+0.082, +0.211]  | MATCH    |
| E3   | variable ruler mask – ERM @50%           | -0.094        | -0.094 [-0.168, -0.021]  | MATCH    |
| E3   | variable ruler mask – ERM @100%          | -0.155        | -0.155 [-0.228, -0.084]  | MATCH    |
| E3   | variable ruler balanced – ERM @0%        | +0.103        | +0.102 [+0.068, +0.140]  | MATCH    |
| E3   | variable ruler balanced – ERM @50%       | +0.171        | +0.171 [+0.132, +0.213]  | MATCH    |
| E3   | variable ruler balanced – ERM @100%      | +0.222        | +0.222 [+0.181, +0.265]  | MATCH    |
| E4   | occlusion (50/50) mask – ERM @0%         | +0.035        | +0.036 [-0.016, +0.086]  | MISMATCH |
| E4   | occlusion (50/50) mask – ERM @50%        | +0.033        | +0.041 [-0.010, +0.090]  | MATCH    |
| E4   | occlusion (50/50) mask – ERM @100%       | +0.039        | +0.039 [-0.010, +0.091]  | MISMATCH |
| E5   | dilate0 – ERM @50%                       | -0.097        | -0.097 [-0.164, -0.031]  | MATCH    |
| E5   | dilate10 – ERM @50%                      | -0.138        | -0.138 [-0.203, -0.074]  | MATCH    |
| E5   | dilate25 – ERM @50%                      | -0.153        | -0.153 [-0.221, -0.089]  | MATCH    |
| E5   | dilate50 – ERM @50%                      | -0.142        | -0.142 [-0.206, -0.083]  | MATCH    |
| E5   | dilate0 – ERM @100%                      | -0.128        | -0.129 [-0.195, -0.066]  | MATCH    |
| E5   | dilate10 – ERM @100%                     | -0.143        | -0.144 [-0.206, -0.082]  | MATCH    |
| E5   | dilate25 – ERM @100%                     | -0.144        | -0.144 [-0.209, -0.082]  | MATCH    |
| E5   | dilate50 – ERM @100%                     | -0.138        | -0.138 [-0.200, -0.082]  | MATCH    |
| E6   | $\Delta p$   erm @100%                   | +0.235        | +0.235                   | MATCH    |
| E6   | $\Delta p$   mask @100%                  | +0.490        | +0.491                   | MATCH    |
| E6   | $\Delta p$   mask @0%                    | +0.000        | +0.000                   | MATCH    |
| E6   | $\Delta p$   erm @0%                     | +0.128        | +0.128                   | MATCH    |
| E6   | $\Delta p$   erm @50%                    | +0.195        | +0.195                   | MATCH    |
| E7   | mask – ERM, large lesions @0%            | +0.249        | +0.249                   | MATCH    |
| E7   | mask – ERM, medium lesions @0%           | +0.214        | +0.215                   | MATCH    |
| E7   | mask – ERM, small lesions @0%            | +0.150        | +0.150                   | MATCH    |
| E7   | mask – ERM, large lesions @50%           | +0.087        | +0.087                   | MATCH    |
| E7   | mask – ERM, medium lesions @50%          | -0.129        | -0.129                   | MATCH    |
| E7   | mask – ERM, small lesions @50%           | -0.325        | -0.326                   | MATCH    |
| E7   | mask – ERM, large lesions @100%          | +0.064        | +0.063                   | MATCH    |
| E7   | mask – ERM, medium lesions @100%         | -0.158        | -0.158                   | MATCH    |
| E7   | mask – ERM, small lesions @100%          | -0.364        | -0.364                   | MATCH    |
| E14  | in-lesion hair median area ratio         | +0.033        | +0.033                   | MATCH    |
| E14  | Trap A mask – ERM, area-ratio tertile T1 | -0.064        | -0.057 [-0.095, -0.022]  | MATCH    |

| Exp.  | Quantity                                      | Pilot (point) | Replication (crossed CI) | Verdict  |
|-------|-----------------------------------------------|---------------|--------------------------|----------|
| E14   | Trap A mask – ERM, area-ratio tertile T2      | -0.057        | -0.046 [-0.082, -0.013]  | MATCH    |
| E14   | Trap A mask – ERM, area-ratio tertile T3      | -0.059        | -0.037 [-0.071, -0.005]  | SIGN+CI  |
| E10   | hair decodability test AUROC                  | +0.943        | +0.943 [+0.919, +0.964]  | MATCH    |
| E10   | detector mean IoU                             | +0.220        | +0.222                   | MATCH    |
| E10   | detector median IoU                           | +0.180        | +0.184                   | MATCH    |
| E10   | LEACE A-decoder after erasure                 | +0.500        | +0.500                   | MATCH    |
| E10   | clean melanoma AUROC after erasure            | +0.814        | +0.815                   | MATCH    |
| E10   | melanoma-head $ \Delta p $ after erasure      | +0.058        | +0.057                   | MATCH    |
| E10   | LEACE projection rank                         | +1.000        | +1.000                   | MATCH    |
| E12   | overlap-contrast trap mask – ERM (spec: null) | +0.047        | -0.103 [-0.121, -0.085]  | MISMATCH |
| E8/E9 | inpaint – ERM @50%                            | +0.211        | +0.211 [+0.175, +0.249]  | MATCH    |
| E8/E9 | inpaint – ERM @100%                           | +0.244        | +0.243 [+0.205, +0.284]  | MATCH    |
| E8/E9 | inpaint_consistency – ERM @50%                | +0.046        | +0.044 [-0.007, +0.092]  | MATCH    |
| E8/E9 | inpaint_consistency_lam0 – ERM @50%           | +0.036        | +0.035 [-0.015, +0.086]  | MATCH    |
| E8/E9 | leace – ERM @100%                             | +0.294        | +0.294 [+0.249, +0.339]  | MATCH    |
| E8/E9 | balanced – ERM @100%                          | +0.281        | +0.281 [+0.236, +0.329]  | MATCH    |
| E8/E9 | dfr – ERM @100%                               | +0.279        | +0.279 [+0.220, +0.339]  | MATCH    |
| E8/E9 | groupdro – ERM @100%                          | +0.197        | +0.197 [+0.155, +0.240]  | MATCH    |
| E13   | trapA mask – ERM (rev)                        | -0.064        | -0.049 [-0.083, -0.018]  | MATCH    |
| E13   | trapB mask – ERM (rev)                        | +0.124        | +0.098 [+0.060, +0.136]  | SIGN+CI  |
| E13   | trapA inpaint – ERM (rev)                     | +0.101        | +0.098 [+0.084, +0.112]  | MATCH    |
| E13   | trapB inpaint – ERM (rev)                     | +0.108        | +0.100 [+0.083, +0.117]  | MATCH    |
| E13   | trapA balanced – ERM (rev)                    | +0.213        | +0.213 [+0.183, +0.243]  | MATCH    |
| E13   | trapB balanced – ERM (rev)                    | +0.234        | +0.204 [+0.166, +0.245]  | SIGN+CI  |
| E13   | trapA dfr – ERM (rev)                         | +0.202        | +0.196 [+0.157, +0.235]  | MATCH    |
| E13   | trapB dfr – ERM (rev)                         | +0.192        | +0.196 [+0.147, +0.251]  | MATCH    |
| E13   | trapA leace_paired – ERM (rev)                | +0.058        | +0.074 [+0.062, +0.086]  | MATCH    |
| E13   | trapB leace_paired – ERM (rev)                | +0.078        | +0.078 [+0.059, +0.097]  | MATCH    |
| E13   | trapA leace_unpaired – ERM (rev)              | +0.316        | +0.320 [+0.280, +0.361]  | MATCH    |
| E13   | trapA mask – ERM, HAM only                    | -0.030        | -0.011 [-0.064, +0.043]  | MISMATCH |
| E13   | trapA mask – ERM, BCN only                    | -0.078        | -0.067 [-0.110, -0.025]  | MATCH    |
| E15   | trapA mask – ERM (rev)                        | -0.069        | -0.104 [-0.143, -0.067]  | SIGN+CI  |
| E15   | trapA balanced – ERM (rev)                    | +0.126        | +0.146 [+0.122, +0.170]  | MATCH    |
| E15   | trapA dfr – ERM (rev)                         | +0.114        | +0.116 [+0.090, +0.145]  | MATCH    |
| E15   | trapA leace_paired – ERM (rev)                | +0.031        | +0.034 [+0.026, +0.042]  | MATCH    |
| E15   | trapA leace_unpaired – ERM (rev)              | +0.192        | +0.203 [+0.165, +0.243]  | MATCH    |

**The mismatches** With the corrected estimator the ledger has 5 mismatches (originally 3). (i) E1, masking at 25% overlap: the pilot’s interval excluded zero, the corrected one includes it (the harm starts later than the pilot said, at 50%). (ii) E4, the occlusion control at 0% and 100% overlap: the pilot’s small positive effects are no longer significant; the conclusion — occlusion by the mask does not harm — is unchanged (Stage 2). (iii) E12, the overlap-contrast trap: null in the pilot, clearly non-null in the replication (Stage 3 explains why this design was abandoned). (iv) One HAM-only stratum of the hair trap. None of them affects the central claims.

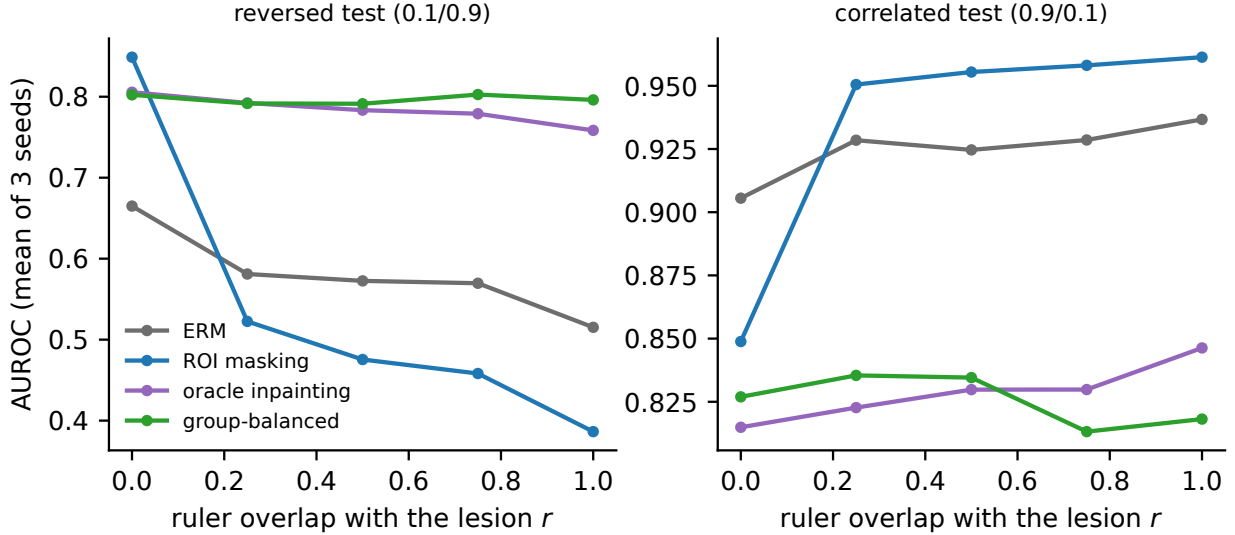


Figure 2: The pilot’s question, regenerated: reversed-test (left) and correlated-test (right) AUROC of the main arms as the ruler moves into the lesion (ISIC 2018, DINOv2 @518). Masking helps most when the ruler lies outside the lesion and falls below ERM once it overlaps it; the masked model’s correlated-test AUROC rises with overlap, which is the shortcut it now relies on.

**Leakage** Random splits leave near-duplicate images on both sides of the split and inflate the measured ERM shortcut gap by 18% (no overlap) and 10% (full overlap), and shift both masking effects (Table 4); every analysis therefore uses grouped splits.

Table 4: Leakage: grouped versus random image-level splits (controlled sweep, DINOv2@518).

| Overlap | Split              | Near-duplicate pairs across split | ERM gap (corr – rev) | mask – ERM (rev) |
|---------|--------------------|-----------------------------------|----------------------|------------------|
| 0%      | grouped            | 0                                 | 0.241                | +0.184           |
|         | random (mean of 5) | 93                                | 0.284                | +0.223           |
| 100%    | grouped            | 0                                 | 0.421                | −0.129           |
|         | random (mean of 5) | 93                                | 0.465                | −0.088           |

## 5.1 Consequences of the bootstrap correction

Across 2,454 regenerated intervals, the corrected intervals are wider by a median factor of 1.55 (interquartile range 1.33–1.74; Figure 3); 146 intervals that excluded zero no longer do and 32 that included zero now exclude it; 160 regenerated point estimates moved by more than 0.03 (largest 0.222, in the capsule synthetic sweeps, whose split was re-dealt when the cohort was rebuilt; Stage 4). The claims that changed are listed in Stage 4 and in `results/bootstrap_correction/sweeps_umte_old_vs_new.md`.

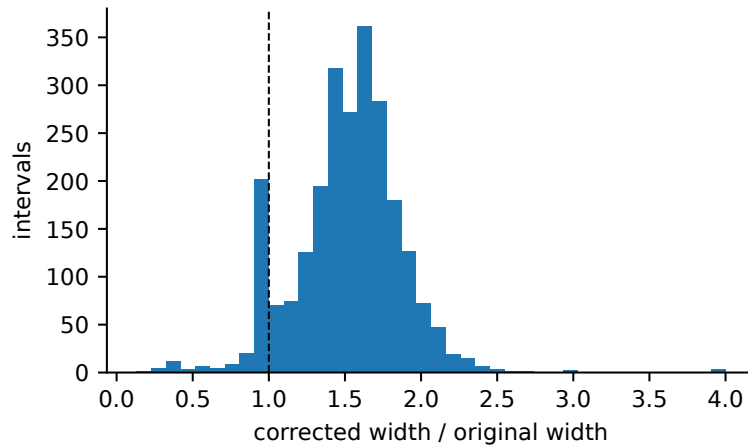


Figure 3: Width of the corrected interval divided by the width of the original one, for every regenerated interval.

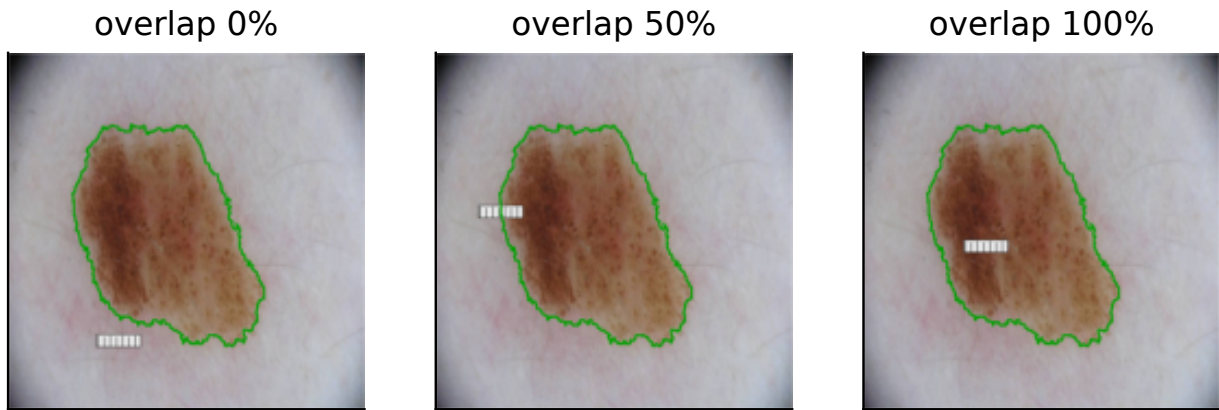


Figure 4: The controlled dermoscopy sweep: the same ISIC 2018 image with the synthetic ruler at 0, 50 and 100% overlap with the lesion (green: lesion mask). ISIC images CC BY-NC 4.0.

## 6 What failed or changed

- **Bootstrap flaw (found in the third review round).** The inherited estimator understated uncertainty whenever seeds share test images. All runs were regenerated and every interval recomputed; claims that lost support are listed in Stage 4.
- **Three pilot mismatches**, above; the E12 design failure is discussed in Stage 3.
- **Rebuild on new hardware.** All data were re-downloaded and every derived object rebuilt on a different GPU; the lesion U-Net was retrained, which changed hair-trap membership by about 3%. Two gaps in the reproduction recipe (`Makefile`) were found and fixed.
- **Commit times are not independent proof** of pre-registration order (above).

## 7 Conclusion

Shortcut artifacts are a clinical safety problem, and ROI masking — the usual defence — rests on an untested assumption about where the shortcut lives. The pilot suggested that masking reverses from helpful to harmful as the artifact moves into the lesion; an independent re-implementation



reproduced it. The protocol — trap environments, frozen foundation-model features, grouped splits, pre-registration and a crossed bootstrap — is fixed here and used unchanged in every later stage. **Supported:** the pilot’s central sweep result and most of its numbers replicate. **Not supported:** the pilot’s null overlap-contrast result (E12); the original, narrower intervals.

## 8 Questions for the supervisor

1. Is the reversed-environment AUROC the right primary metric for the thesis, or should the thesis lead with a clinical metric (sensitivity at a fixed specificity, Stage 5)?
2. How should the thesis present the bootstrap correction: as a methods footnote, or as a contribution (it affects any study that averages several training seeds on one test set)?
3. Is it acceptable that pre-registrations before the last rounds carry only local commit times?

## 9 Next stage

Stage 2 returns to the controlled dermoscopy sweep and asks *why* masking reverses: it separates occlusion, lost context and retention of the artifact with dedicated experiments (occlusion control, mask dilation, counterfactual sensitivity, lesion-size strata), repeats the sweep in ultrasound, capsule endoscopy and chest radiography, and shows all of it with the corrected intervals.

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